Daniel **Carrión**

Engineer & Designer

About me

Electronics, Robotics and Mechatronics Engineer with a Master's degree in Mechatronics. I work in R&D and cybersecurity applied to embedded systems and IoT, with a focus on firmware (C/C++, FreeRTOS/ESP32), instrumentation, sensing and test automation (Python). I have developed infrastructures and applications for wireless sensor networks and edge computing in projects such as HortSost, gaining deeper expertise in architectures like RISC-V.

Personal

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DaniCampusIoT
danrodcar
danicampusiot.github.io

Areas of specialization

C/C++ Programming • IoT • PCB Design • 3D Design • Electronic • Robotic & Control

Skills

Coding and Programming
Data Analytics
Data interpretation and Charting
Active listening and Empathy
Adaptability
Continuous learning

Interests

SoC Programming / Automating processes Web Programming / Reverse engineering / Discuss

Languages

Spanish C2 mother tongue
English B2 FCE Cambridge

Short Resumé

01/2025–02/2026

Test Engineer & Certification Specialist

IoT, Cybersecurity & Compliance · DEKRA

- Member of the evaluation team for Google pKVM (Android) SESIP Level 5 certification: cryptographic analysis, secure virtualization design review and preparation of assessment evidence.
- Security assessment of IoT devices and interfaces (Bluetooth Classic/BLE/Mesh, LTE, Ethernet/MACsec, I2C, JTAG): attack surface analysis, pairing, encryption/key management, domain isolation and Layer 2 traffic protection.
- Analysis and verification of critical security features (secure boot, secure firmware upgrade, secure debug, secure storage): design/implementation validation and resilience against software and physical attacks.
- Daily development and execution of test campaigns in Linux/VM environments (bash/CMD): log analysis, test environment management and automation using in-house tooling (Python and C/C++).

02/2022–09/2024

Technical staff & Research personnel

Programmer · COMPUTERS ARCHITECTURE DEPT., UMA

- Software development and technical support for the HortSost II project (University of Málaga): technical adaptation, solution integration and maintenance.
- Technical lead for greenhouse automation: instrumentation and sensing, electrical setup, ESP32 firmware and data routing/management via MQTT and Node-RED on UNIX servers.
- Development of a self-configuring sensor infrastructure for edge computing: C/C++, FreeRTOS and ESP32 for data acquisition and edge processing.
- Research and knowledge transfer: publication and talk "Self-configuring wireless sensor network for edge computing" at a national computer science conference; signal conditioning for the A121 (60 GHz PCR) radar sensor for indoor biometric monitoring.
- Co-supervision of undergraduate thesis projects, including a project awarded 2nd place in the Bizintek Awards 22/23.

06/2019–01/2020

Technical Staff

Worker · SITER 2000

- Internship focused on consulting and studies of GSM/DCS/UMTS and FTTH mobile network deployments.
- Analyzed engineering documents and As-builts for access point installations in educational centers.

Degrees

2023 **Master in Mechatronics Engineering**
M.A. · University of Malaga

2023 **Project Management of Photovoltaic Solar Installations: Solar Project Management**
M.A. · EUROINNOVA BUSINESS SCHOOL

2021 **Grad. Electronic engineering, robotics and mechatronics**
B.E. · University of Seville-Malaga

Certificates

2024 **CATIA V5 - Part Design (Ass)**
Cert. · Dassault Systèmes

2020 **PHP, JAVASCRIPT and MySQL developer**
Cert. · DEUSTO FORMACIÓN

2019 **RPAS pilot and radio operator**
Cert. · Aerocamaras

Publication & Talks

Jun. 2024 "Auto-configurable wireless sensor network for edge computing", at: *Spanish Congress of Informatics* in A Coruña, Jun. 2024.